

CSCN8020 Presentation Transcript: Slides 1 to 4

Presenter: Ali Cihan Ozdemir | **Course:** CSCN8020 - Reinforcement Learning | **Team:** Group 4

Target Time per Slide: 50 Seconds (~110 words) | **Total Time:** 3 Minutes 20 Seconds

■■ SLIDE 1: Title & Project Overview

■■ Target: 50s | ■ Word Count: 110 words | ■■ Action: Point to title & team names

Spoken Transcript:

Good morning Professor Espinosa and classmates. I'm Ali Cihan Ozdemir, and together with my team members Lohith, Sumanth, and Muthuraj, we represent Group 4.

*Today, we're taking reinforcement learning beyond basic single-joint control to tackle a complex, high-dimensional robotics challenge: **Namaste-G1: Teaching a Humanoid Robot to Greet.***

Our main goal is to train a simulated 14-degree-of-freedom Unitree G1 humanoid robot to execute a full two-arm Namaste greeting pose—reaching the pose, holding it steadily within strict joint tolerances, and returning gracefully to rest.

Here is how we designed and evaluated multi-joint humanoid control here at Conestoga College.

■■ SLIDE 2: Today's Plan (Agenda)

■■ Target: 50s | ■ Word Count: 110 words | ■■ Action: Point to roadmap items

Spoken Transcript:

Here is the roadmap for our presentation today.

We will begin by reviewing our initial 14-joint Dueling DQN baseline, and explain why we transitioned from discrete step control to continuous Actor-Critic.

Next, we will reveal our major headline finding and the critical PyTorch gradient bug we discovered along the way.

We will then analyze what happened after fixing the bug—where the robot demonstrated real learning movement but experienced policy instability—followed by explaining the underlying mathematical principles in plain terms, running a live simulation demonstration, and detailing our next steps for the project.

■■ SLIDE 3: Before Actor-Critic: The DQN Baseline

■■ Target: 50s | ■ Word Count: 110 words | ■■ Action: Point to 14-joint stats (3.1 to 5.2 joints)

Spoken Transcript:

To establish our baseline, we first designed a 14-joint branching Dueling Q-Network. Each joint had three discrete actions: hold the current position, decrease the angle, or increase the angle.

We ran a fully logged 200-episode diagnostic experiment to evaluate its learning capability. While the network successfully learned to increase the average number of joints in tolerance from 3.1 up to 5.2—reaching a peak moment where 7 out of 14 joints met the criteria—it achieved zero full 14-joint successes across the entire run.

This proved that coarse, stepped discrete control made fine multi-joint coordination extremely difficult.

■■ SLIDE 4: Why We Moved to Actor-Critic

■■ Target: 50s | ■ Word Count: 110 words | ■■ Action: Point to DQN Stepped vs Actor-Critic Smooth

Spoken Transcript:

That key limitation directly motivated our shift to Actor-Critic.

While DQN could only move each joint in fixed, stepped increments, Actor-Critic outputs a continuous action value for every joint simultaneously. Achieving finer, smoother control is essential when coordinating 14 joints at the exact same time.

However, moving to continuous control significantly expanded the search space, making the task harder rather than easier.

Consequently, our refined goal expanded into three distinct phases: reaching the Namaste pose, holding it steadily for a consecutive streak, and returning smoothly back to a resting posture.